Assignment 2

1) Autoencoder vs classifier

- Classifier: predicts labels.
- Autoencoder: Compress input data ,reconstructs input.
  Training signal = **input itself**.

2) Bottleneck

The bottleneck is:The low-dimensional latent representation z.

It forces the model to:

- Compress information
- Keep only important features
  If same size as input ,Then the model could:
- Simply copy the input directly
- Memorize instead of learning meaningful representations

The autoencoder would become almost useless because no compression happens.

3) Why random z gives noise in plain AE

Latent space is **unstructured**.

During training:

- Encoder maps inputs to arbitrary regions in latent space.
- Only some regions contain meaningful encoded faces.

the sampled point likely lands in an unseen region never learn how to decode them ,So the output is noise .

4) How VAE fixes it

A VAE forces latent vectors to follow a known distribution.

Encoder outputs **mean and variance** → latent space becomes **smooth and continuous**.

5) Reparameterization trick

Problem: Sampling is random and non-differentiable.

Backpropagation cannot pass gradients through random sampling directly.

Rewrite sampling as: z is a deterministic function of μ, σ, plus a random noise ε that does not depend on any parameters.
Allows gradient backpropagation through randomness.

6) Generating from VAE

1. Sample latent vector:z
2. Pass z to decoder to generate the image .
   Works because VAE forces latent space to follow Gaussian distribution.

7) Reconstruction loss

Original: [0.8, 0.2, 0.6, 0.4]

(a) First reconstruction
(0.7-0.8)^2 + (0.3-0.2)^2 + (0.5-0.6)^2 + (0.5-0.4)^2 = 0.01+0.01+0.01+0.01 = 0.04

(b) Second reconstruction

(0.6-0.8)^2+(0.4-0.2)^2+(0.4-0.6)^2+(0.6-0.4)^2 = 0.04+0.04+0.04+0.04 = 0.16
Better reconstruction = **first one**.

(c) What prevents large z?

KL divergence regularization.

KL term forces latent distributions to stay close to standard Gaussian.

Programming answers

Structured sampling gives better images
Because latent grid explores learned manifold.

Random points → may fall in low-probability regions.

latent_dim 2 → 8

- Reconstructions improve.
- Harder to visualize latent space.

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